MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 01
Chapter 13: Statistics

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 01

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. For a given frequency distribution, the class mark of the class interval 15 – 35 is: [1]

(A) 20 (B) 25 (C) 30 (D) 15

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 25

Justification:

$$\text{Class Mark} = \frac{\text{Upper Limit} + \text{Lower Limit}}{2} = \frac{35 + 15}{2} = \frac{50}{2} = \mathbf{25}.$$

$$\boxed{\text{Class Mark} = \mathbf{25}}$$

Q2. While computing the mean of grouped data, it is assumed that the frequencies are: [1]

(A) centred at the upper limits of the classes
(B) centred at the lower limits of the classes
(C) centred at the class marks of the classes
(D) evenly distributed across all classes

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) centred at the class marks of the classes

Justification: In calculating the mean of grouped data, all values within a class interval are assumed to be concentrated at the midpoint or class mark (x_i) of that class.

Q3. If $\sum f_i = 20$ and $\sum f_i x_i = 200 + 5k$, and the mean of the distribution is 15, then the value of k is: [1]

(A) 15 (B) 20 (C) 25 (D) 30

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 20

Justification:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \implies 15 = \frac{200 + 5k}{20}$$

$$15 \times 20 = 200 + 5k \implies 300 = 200 + 5k \implies 5k = 100 \implies \boxed{k = \mathbf{20}}.$$

Q4. The modal class for the following distribution is: [1]

Class	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency	7	14	13	12	20

(A) $10-20$ (B) $20-30$ (C) $30-40$ (D) $40-50$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $40-50$

Justification: The modal class is the class interval having the maximum frequency. Here, the highest frequency is 20, which corresponds to the class interval $\mathbf{40-50}$.

Q5. **Assertion (A):** The mean of the first 10 prime numbers is 12.9.
Reason (R): The first 10 prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: The first 10 prime numbers are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$. Hence, Reason (R) is true.
Sum of these 10 primes $= 2+3+5+7+11+13+17+19+23+29 = 129$.

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}} = \frac{129}{10} = 12.9.$$

Thus, Assertion (A) is true, and Reason (R) correctly explains Assertion (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. Find the mean of the first 6 multiples of 3. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Listing Multiples and Calculating Sum [1 Mark]

The first 6 multiples of 3 are: $3, 6, 9, 12, 15, 18$.
Total number of observations $n = 6$.

$$\text{Sum of observations } (\sum x_i) = 3 + 6 + 9 + 12 + 15 + 18 = 63.$$

Step 2: Formula Declaration & Mean Calculation [1 Mark]

$$\text{Mean } (\bar{x}) = \frac{\sum x_i}{n} = \frac{63}{6} = \frac{21}{2} = 10.5.$$

Mean of the first 6 multiples of 3 = **10.5**

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. Find the arithmetic mean of the following distribution using the Direct Method: [3]

Class Interval	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency (f_i)	7	10	15	8	10

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Direct Method Calculation Table [1.5 Marks]

Class Interval	Frequency (f_i)	Class Mark (x_i)	$f_i x_i$
$0-10$	7	5	35
$10-20$	10	15	150
$20-30$	15	25	375
$30-40$	8	35	280
$40-50$	10	45	450
Total	$\sum f_i = 50$		$\sum f_i x_i = 1290$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1290}{50} = \frac{129}{5} = 25.8$$

Step 3: Final Boxed Result [0.5 Mark]

Arithmetic Mean $(\bar{x})$ = **25.8**

Q8. Find the mode of the following grouped frequency distribution: [3]

Class	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$
Frequency	4	8	10	12	6	4

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying Modal Class and Parameters [1 Mark]

The maximum frequency is 12, which lies in the class interval **40 – 50**.
∴ Modal Class = 40 – 50.
Parameters:

- Lower limit of modal class $(l) = 40$
- Frequency of modal class $(f_1) = 12$
- Frequency of preceding class $(f_0) = 10$
- Frequency of succeeding class $(f_2) = 6$
- Class width $(h) = 50 - 40 = 10$

Step 2: Mode Formula & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 40 + \left(\frac{12 - 10}{2(12) - 10 - 6}\right) \times 10 = 40 + \left(\frac{2}{24 - 16}\right) \times 10 = 40 + \left(\frac{2}{8}\right) \times 10$$

Step 3: Calculation & Final Result [1 Mark]

$$\text{Mode} = 40 + \frac{20}{8} = 40 + 2.5 = 42.5$$

$$\boxed{\text{Mode of the distribution} = \mathbf{42.5}}$$

Q9. Compute the median of the following frequency distribution: **[3]**

Class Interval	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120
Frequency	6	8	10	12	6	5

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
0 – 20	6	6
20 – 40	8	6 + 8 = 14
40 – 60	10	14 + 10 = 24
60 – 80	12	24 + 12 = 36
80 – 100	6	36 + 6 = 42
100 – 120	5	42 + 5 = 47
Total	$N = \sum f_i = 47$	

Step 2: Median Class Identification & Parameters [1 Mark]

Here, $N = 47 \implies \frac{N}{2} = \frac{47}{2} = 23.5$.
The cumulative frequency just greater than 23.5 is 24, which corresponds to the class interval **40 – 60**.
∴ Median Class = 40 – 60.
Lower limit $(l) = 40$, Cumulative frequency of preceding class $(cf) = 14$,
Frequency of median class $(f) = 10$, Class width $(h) = 20$.

Step 3: Formula & Calculation [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$$

$$\text{Median} = 40 + \left(\frac{23.5 - 14}{10}\right) \times 20 = 40 + \left(\frac{9.5}{10}\right) \times 20 = 40 + 9.5 \times 2 = 40 + 19 = 59$$

$$\boxed{\text{Median of the distribution} = \mathbf{59}}$$

Q10. CASE STUDY: HOSPITAL INPATIENT COVID-19 CLINICAL ADMISSIONS PROFILE [4 Marks]

A municipal medical hospital recorded the age distribution of patients admitted during a seasonal influenza and respiratory illness surge to allocate medical oxygen, intensive care beds, and nursing personnel efficiently.

Age Interval (years)	$5-15$	$15-25$	$25-35$	$35-45$	$45-55$	$55-65$
Number of Patients	6	11	21	23	14	5

(i) Identify the modal class of the admitted patients and state its lower limit. [1]

(ii) Calculate the mode of the age distribution of the patients. [2]

(iii) Find the class mark and cumulative frequency of the modal class. [1]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Subpart (i): Modal Class & Lower Limit [1 Mark]

The highest number of patients (frequency) is 23, which falls in the age group $\mathbf{35-45}$.

$\therefore$ Modal Class $= 35-45$.

Lower limit of the modal class $(l) = 35$.

$$\boxed{\text{Modal Class} = \mathbf{35-45}, \quad \text{Lower Limit } (l) = \mathbf{35}}$$

Subpart (ii): Mode Calculation [2 Marks]

Here: $l = 35$, $f_1 = 23$, $f_0 = 21$ (preceding frequency), $f_2 = 14$ (succeeding frequency), $h = 10$.

Formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{ [1 Mark]}$$

$$\text{Mode} = 35 + \left(\frac{23-21}{2(23)-21-14}\right) \times 10 = 35 + \left(\frac{2}{46-35}\right) \times 10 = 35 + \frac{20}{11}$$

$$\text{Mode} = 35 + 1.818... \approx 36.82 \text{ years} \quad \left(\text{or } 36\frac{9}{11} \text{ years}\right) \text{ [1 Mark]}$$

$$\boxed{\text{Mode of the distribution} = \mathbf{36.82} \text{ years} \quad \left(\text{or } \mathbf{36\frac{9}{11}} \text{ years}\right)}$$

Subpart (iii): Class Mark & Cumulative Frequency of Modal Class [1 Mark]

For the modal class $35-45$:

$$\text{Class Mark } (x_i) = \frac{35+45}{2} = \frac{80}{2} = \mathbf{40}.$$

Cumulative frequency up to the modal class:

$$cf = 6 + 11 + 21 + 23 = \mathbf{61}.$$

$$\boxed{\text{Class Mark} = \mathbf{40}, \quad \text{Cumulative Frequency} = \mathbf{61}}$$

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